James Akl

Researcher & Engineer

Robotics, AI, Control, and Data

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Pitch: I am structured, rigorous, and well-versed in formal mathematics, software, and robotic systems. I seek to create value through helping to develop and deploy autonomous and intelligent systems.

COURSEWORK

WPI Transcript | LAU Transcript

Select coursework is tabulated below. Transcripts are linked above to view all courses. Codes are listed to verify courses on transcript.

Abbreviations GR = *Graduate* UG = *Undergraduate* WPI = *Worcester Polytechnic Institute* LAU = *Lebanese American University*

TOPIC GROUP	TOPIC	LEVEL	INSTITUTION	GRADE	COURSE CODES
Robotics & AI	Foundations of Robotics	GR	WPI	A	RBE 500
	Artificial Intelligence	GR	WPI	A	CS 534
	Robot Dynamics	GR	WPI	A	RBE 501
	Robot Control	GR	WPI	A	RBE 502
	Biomechanics & Robotics	GR	WPI	A	RBE 520
	Robot/AI Ethics & Future of Work	GR	WPI	A	WR 593
Mathematics *Pure*	Convex Analysis & Optimization	GR	LAU	A	ACM 720
	Real Analysis	UG	LAU	A	MTH 401
	Discrete Structures	UG	LAU	A	MTH 207
Mathematics *Applied*	Computational Inverse Problems	GR	WPI	A	MA 590
	Applied Statistics	GR	WPI	A	MA 511
	Engineering Mathematics	GR	WPI	A	MA 501
	Numerical Linear Algebra	GR	LAU	A	ACM 713
	Numerical Methods	UG	LAU	A	GNE 333
	Probability & Statistics	UG	LAU	A	GNE 331
	Multivariate Calculus (4 courses)	UG	LAU	A	MTH 101, 102, 201, 206
	Differential Equations	UG	LAU	A	MTH 304
	Finite Element Methods	UG	LAU	A	MEE 421
	Optimization	UG	LAU	A	INE 420
Computer Science	Quantum Cryptography	GR	WPI	A	ECE 673
	Data Structures & Algorithms	GR	WPI	A	CS 5007
	Object-Oriented Programming	UG	LAU	A	COE 212
	MATLAB Programming	UG	LAU	A	MEE 212
Management *Business/Economics*	Entrepreneurship & Innovation	GR	WPI	A	ETR 500
	Engineering Economy	UG	LAU	A	INE 320
	Microeconomics	UG	LAU	A	ECO 201
	Project Management	UG	LAU	B	INE 428
Engineering *General*	Control Systems	UG	LAU	A	MEE 445, 446
	Instrumentation & Measurements	UG	LAU	B	MEE 391
	Electrical Circuits	UG	LAU	B	ELE 305
	Computer Aided Design	UG	LAU	A	MEE 351
	Engineering Graphics	UG	LAU	A	MEE 211
Engineering *Mechanical*	Dynamics & Statics (3 courses)	UG	LAU	A	PHY 111, CIE 200, MEE 241
	Mechanical Vibrations	UG	LAU	A	MEE 442, 443
	Kinematics of Linkages	UG	LAU	B	MEE 341
	Mechanical Design	UG	LAU	B	MEE 422
Engineering *Thermal*	Fluid Mechanics	UG	LAU	B	MEE 311, 312
	Thermodynamics (2 courses)	UG	LAU	B	MEE 301, 302
	Heat Transfer	UG	LAU	B	MEE 403, 404
	Internal Combustion Engines	UG	LAU	A	MEE 407
	Refrigeration & Air Conditioning	UG	LAU	B	MEE 515, 516
	Thermal System Design	UG	LAU	B	MEE 414
Engineering *Materials*	General Chemistry (2 courses)	UG	LAU	A	CHM 101, 200
	Material Properties	UG	LAU	A	MEE 321
	Strength of Materials	UG	LAU	B	MEE 320
	Manufacturing Processes	UG	LAU	B	MEE 332, 333